Training-Consistent Pressure-Prior Integration for Independent Multi-Agent Traffic Signal Control: Confirmatory Evidence and Setting-Dependent Boundaries

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Abstract

Pressure is the organizing quantity of classical traffic signal control and of several reinforcement learning (RL) formulations; in the published pressure-informed RL controllers we are aware of, however, pressure enters mainly through the reward or the observation of value-based agents. This study evaluates a different integration: a pressure-based action prior (a per-intersection preference for signal phases that relieve queue imbalance, added to the policy logits) given to independent proximal policy optimization (PPO) agents and applied consistently during both rollout sampling and PPO updates. Under an internally pre-registered, hash-locked protocol frozen before data collection, the resulting method, IPPO-P, was compared with an otherwise identical IPPO baseline in three synthetic grid scenarios spanning two network topologies (a 3×3 grid and a 2×2 grid that carries two demand levels), using ten new independent training seeds, five deterministic evaluation episodes per task in fresh environments, and one pre-registered primary outcome: demand service time (U; lower is better), the mean per-vehicle service time accrued by all planned demand vehicles, with unfinished vehicles censored at the demand horizon. All 60 planned tasks completed, with the demand-state bookkeeping check satisfied and zero teleport events across the 300 evaluation episodes. IPPO-P improved U by 1.712% (seed-level log-ratio effect; bootstrap 95% CI [1.252%, 2.202%], 20,000 paired resamples), with all ten training-seed effects positive and an exact two-sided paired sign-flip p = 0.001953, roughly 0.5–4.5 s of service time per vehicle depending on scenario. Scenario-specific effects (+0.407%, +1.023%, +3.675%) were secondary and descriptive. Post-confirmation exploratory scale-ups showed that the gain is not universal: the effect was positive at 2700 veh/h (+0.458%, n = 5), negative at 5400 veh/h (−0.997%, 0/5 seeds positive, n = 5), and weakly positive at 6300 veh/h (n = 2, one near-tie). Descriptive non-learning anchors bound the scale of the comparisons: both learned policies sit well below a fixed-time program, and IPPO-P's mean U lies 2.9%–6.9% below a max-pressure rule that in turn edges IPPO in the heaviest scenario; these figures give context on scale, not ranked claims. Within this IPPO-internal controlled evaluation, at a single frozen prior scale, training-consistent pressure-prior integration improved demand service time relative to matched IPPO, while setting-dependent boundaries caution against extrapolation.

Index Terms

traffic signal control; multi-agent reinforcement learning; independent proximal policy optimization; action prior; internally pre-registered confirmatory evaluation; demand service time

A. Introduction

Traffic signal control (TSC) is a sequential decision problem in which locally acting controllers jointly determine network-level delay: each controller's best action depends on the queue states that neighboring controllers are simultaneously shaping, so independent adaptation makes the environment non-stationary from every agent's perspective. Stable policy learning under this coupling, whether through explicit coordination mechanisms, architectural inductive bias, or training-stability safeguards, is a central concern of the field. Existing practice spans fixed-time plans, actuated rules, and learning-based controllers [1]. Multi-agent reinforcement learning (MARL) entered the field early [2], and deep RL subsequently produced diagnostic single-intersection agents [3], phase-competition representations [4], parameter-shared decentralized policies for large networks [5], and graph-based network-level cooperation [6]. These mechanisms trade off decentralization against coordination assumptions in different ways. Across the learning-based designs, policies are trained from reward signals; the traffic state, rather than any structured preference over actions, is what the learning process converts into behavior.

A parallel line of work assigns a principled role to pressure. Backpressure scheduling stabilizes constrained queueing networks with maximum-throughput guarantees [7], and max-pressure control extends this idea to signalized intersection networks with well-characterized stability properties [8]. In deep RL for TSC, pressure has been used mainly as

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a learning signal: PressLight trains agents to minimize network pressure through the reward [9]; the PRCOL metric introduced in PDLight refines the pressure definition with downstream lane capacity and enters reward and action logic [10]; and multi-hop upstream pressure generalizes the quantity beyond adjacent links as an RL reward [11]. Reported pressure-informed RL controllers in this line are largely DQN-family agents with pressure in the reward or the state; the pressure signal does not act directly on the sampling distribution of an on-policy policy.

Proximal policy optimization (PPO) [12] is a widely used on-policy algorithm with clipped-ratio updates. Independent PPO (IPPO), in which each agent learns from its own observations without communication or a shared critic, proved unexpectedly strong in the StarCraft multi-agent challenge [13], while MAPPO demonstrated the value of centralized value functions in cooperative games [14]. For TSC, independent PPO matches the deployment-relevant setting of decentralized execution on local observations. Its policy update, however, contains no representation of the traffic-engineering preference that pressure encodes: nothing in the PPO objective distinguishes, a priori, phases that relieve queue imbalance from phases that do not. Whether adding a pressure-derived prior to the policy distribution itself (and doing so consistently during training) improves service quality is an empirical question that the literature reviewed above does not answer.

The empirical TSC-RL literature has grown rapidly [15], and reported gains are sensitive to protocol choices and random-seed variation [16], [17], [18]. Deep RL results vary across training runs, data orders, and stochastic sources [16], [18], and small-sample comparisons can distort effect sizes [17]; single-seed or single-scenario comparisons therefore carry limited evidential weight. This paper is deliberately structured as a confirmatory evaluation: the protocol, the primary contrast, the metric, and the analysis were frozen internally, under a hash-locked, author-controlled protocol (Section 4.1), before the confirmatory data were generated, and the results are reported with seed-level transparency, including settings where the intervention does not help.

The intervention studied here is narrow. IPPO-P (pressure-prior IPPO) adds a per-intersection, per-phase pressure vector to the actor logits before categorical sampling; the vector is centered and standardized within each intersection, clipped to $[-3, 3]$ (a bound that never binds at the tested four-phase intersections; Section 3.3), and scaled by 2.25. The same observation, prior vector, temperature, sampled action, and old log-probability are used in both rollout and PPO update. Nothing else changes: reward, observations, network architecture, and training budget are identical to the IPPO baseline. The contributions are limited to what the data establish:

1. a precisely specified, reproducible IPPO-P implementation with rollout/update distribution consistency (stored-prior reuse, specified at tuple level in Section 3.4);
2. an internally pre-registered (hash-locked), seed-level confirmatory comparison against IPPO over three scenarios (H1–H3; two grid topologies) with ten new independent training seeds;
3. an explicit report of setting-specific boundary evidence from post-confirmation exploratory extensions, including a negative result at 5400 veh/h and an underpowered probe at 6300 veh/h.

The research question is: does incorporating a pressure-based action prior consistently into both rollout action selection and PPO updates improve demand service time over standard IPPO under the tested multi-intersection scenarios, and how does the observed effect vary in exploratory settings? The confirmatory hypothesis is that, across H1, H2, and H3 with equal scenario weights, IPPO-P reduces demand service time relative to IPPO over ten new independent training seeds. Exploratory questions address scenario-specific effect variation, generalization to 3×3 scale-up settings, and, strictly as a hypothesis, a variance-stabilization account of the prior's effect.

This paper proposes no new coordination architecture, pressure metric, or learning algorithm; every component except the logit prior is standard. Whatever value it has lies in the quality of the comparison: one frozen protocol, one internally pre-registered contrast, ten paired seeds, integrity audits on every episode, and an explicit exploratory annex whose negative results are reported with the same care as the positive one. Whether a domain prior belongs inside the training distribution of an on-policy controller is the kind of question that single-run comparisons cannot settle and that the confirmatory design here is built to answer.

Section 2 relates this work to prior literature, Section 3 specifies the method, Section 4 the experimental design and statistical analysis, Section 5 reports results, Section 6 discusses findings and limitations, and Section 7 concludes.

B. Related Work

1) Reinforcement learning for traffic signal control: Multi-agent RL was applied to traffic lights well before the deep learning era, with agents learning value functions from local delay feedback [2]. The classical control stack (fixed-time plans and actuated rules) remains the deployed default against which learning methods are judged [1]. Deep RL then reshaped the field: IntelliLight systematized reward and pipeline design for a single intersection [3]; FRAP modeled the competition between signal phases as a learnable structure [4]; MPLight introduced parameter sharing so that one policy scales across hundreds of decentralized intersections [5]; and CoLight used graph attention over neighboring agents to learn network-level cooperation [6]. These mechanisms occupy different points on the coordination spectrum: parameter sharing reuses capacity without coupling execution, while graph communication couples agents through

learned message passing at both training and execution time. The survey literature organizes the resulting design space by state representation, reward construction, and coordination mechanism [1], [15]. Two features of it motivate the present design. First, evaluation protocols vary widely across studies in simulator choice, demand generation, metric definitions, and the number of random seeds, so cross-paper effect comparisons are unreliable. Second, decentralized policy learning remains an attractive deployment target because it requires neither inter-agent communication nor a centralized coordinator at execution time.

2) Independent and multi-agent PPO: PPO optimizes a clipped-surrogate objective with stable on-policy updates [12]. In the StarCraft multi-agent challenge, IPPO (fully independent actors and critics, no communication) matched or exceeded cooperative methods [13], while MAPPO showed that a centralized value function improves learning in cooperative games [14]. The two results frame a design space: centralized training with decentralized execution relaxes non-stationarity through a global value function at training time, but it presumes access to global state during training and couples agents through the critic, whereas independent PPO needs neither at the cost of treating co-learning agents as part of the environment. For TSC, the independent end of this space is the conservative choice: intersections act on local observations, and no assumption about coordination structure is embedded in the learner. IPPO is therefore a meaningful decentralized baseline, but one with no mechanism to express traffic-domain preferences; that gap is what this paper probes. The question is whether a domain prior, injected into the action distribution itself and kept consistent between training and inference, changes measured service quality under otherwise identical conditions; whether IPPO improves with more capacity or different hyperparameters is a separate question that this design does not touch.

3) Pressure-based control and pressure-informed policies: Pressure, the imbalance between upstream and downstream queues, originated in backpressure scheduling of queueing networks, where it yields maximum-throughput stability [7], and was established for signalized networks as max-pressure control [8], which switches to the phase with maximal net pressure drop under conditions on the network and demand. Within deep RL, PressLight demonstrated that pressure minimization can serve as a reward for coordinated arterial control [9]. PDLight's PRCOL metric improved the pressure definition by accounting for the remaining capacity of outgoing lanes and used it in reward and action logic [10]. More recently, multi-hop upstream pressure extends the quantity beyond adjacent links and is used as an RL reward with anticipatory control [11]. Across this lineage the pressure quantity has been progressively refined, from raw queue imbalance [9] to capacity-aware pressure [10] to multi-hop anticipatory pressure [11], while the agents reported in this lineage are value-based (DQN-type), with pressure shaping rewards or states rather than the sampling distribution of an on-policy policy; we state this as a characterization of the reviewed literature [9]–[11], not as an exhaustive field-level claim.

4) Positioning: Four distinctions locate the present work. First, classical max-pressure control is a deterministic control rule with stability analysis [8], not a learned policy; no comparison to it is claimed here. Second, pressure-reward shaping changes the optimization objective [9], [10]; here the reward is held fixed and identical across methods, so the prior's effect is not confounded with a changed objective. Third, pressure can enter as auxiliary state [6], [11]; here the observation is unchanged, and the prior acts additively on the policy logits. Fourth, biasing actions only at evaluation time would alter the sampling distribution without altering training and would break the importance-ratio semantics of PPO; the intervention evaluated here is instead training-consistent, applying the same modified distribution during rollout and update (Section 3.4). Two further comparison hazards deserve note. An effect attributed to a "prior" may actually originate in accompanying changes to reward scale, state dimension, or network capacity; the design here holds those fixed. And in on-policy methods the sampling distribution is itself part of the training algorithm, so modifying it only at inference changes the object being evaluated, not the object that was trained. This is an evaluation of treatment consistency under an internally pre-registered (hash-locked) protocol, positioned after established pressure-informed prior work [9], [10], [11]; no novelty-priority claim over pressure-based RL is made.

C. Method

1) Problem formulation: Experiments use the SUMO microscopic traffic simulator [19]. All simulator settings not stated below are SUMO defaults: the internal step length is 1 s, vehicles use the default passenger vehicle type, and the teleport mechanism is left at its default (`--time-to-teleport 1000`), so the zero-teleport outcome reported in Section 5 is an empirical integrity gate rather than a configuration guarantee. Each signalized intersection is an agent. The agent's observation concatenates per-lane density over its controlled lanes (vehicle count normalized by lane capacity at a 7.5 m minimum gap) and a one-hot encoding of the current green phase. The action space is the index set of admissible green phases; a 2 s yellow interval is enforced between conflicting green phases (shorter than the 3–5 s typical of urban geometry; the value is fixed and identical across methods and was not varied). Agents act on a 10 s decision interval, and each episode spans a 10,000 s demand horizon during which vehicles enter at fixed route-flow rates (Table 1). A phase change requested by an agent takes effect only when the signal is outside its yellow interval and the current phase has expired, so decision opportunities vary across intersections within a step; the transition process is semi-Markov at the agent level.

The primary outcome is the demand service time U, defined over all planned demand vehicles as the mean of max(0, service_end − scheduled_departure). A vehicle that arrives normally within the demand horizon contributes its arrival time, recorded on the 10 s simulation-step grid (an arrival inside a step is credited to the closing step boundary) and therefore never beyond the horizon; the cap at the horizon end thus binds only at the boundary itself. Every other planned vehicle (not yet inserted, still in the network, temporarily out, or abnormally arrived) is censored at the horizon end. U is therefore not the mean travel time of completed vehicles: unfinished demand accrues the maximal service time rather than being excluded. This censoring makes U sensitive to how a policy treats demand near the end of the horizon and resistant to completion-count inflation, which can flatter policies that finish easy trips while stranding others. Units are seconds; lower is better. The training reward, a cooperative combination of locally normalized speed, queue, and waiting components with a global term weighted 0.30, is an evaluation-independent objective and is identical for both methods; U is not the reward being optimized.

2) IPPO baseline and shared experimental controls: Each agent has an independent actor and an independent critic, both two-layer multilayer perceptrons (128 units, tanh activations). Training uses $\gamma = 0.99$, GAE $\lambda = 0.95$, ratio clip 0.20, Adam with learning rate 3×10^{-4}, entropy coefficient 0.01, value coefficient 0.5, five update epochs, and gradient-norm clipping at 0.5. Advantages are computed with generalized advantage estimation from each agent's own value function, trained on the cooperative reward stream. Episodes end by simulation-time truncation at 10,000 s rather than at a terminal state; the advantage estimate therefore bootstraps with a zero value at the truncated final step (a truncation bootstrap rather than a terminal-state bootstrap), and $\gamma = 0.99$ over 10 s steps attenuates credit by a factor of e over roughly 100 steps (≈ 1,000 s of simulation time). Per-episode training logs (one record per training episode) are retained for every task and included in the data deposit, so learning dynamics are reconstructable, though they are not analyzed in this paper. There is no inter-agent communication, no parameter sharing, and no centralized critic. The IPPO-P critic is structurally identical to the IPPO critic and receives the same local observation: the prior never enters the critic, the reward, or the observation. IPPO and IPPO-P share the environment configuration, reward, training schedule, random-seed structure, and evaluation procedure; the only difference is the action prior (scale 0.0 for IPPO, 2.25 for IPPO-P, fixed in the frozen method lock).

3) Pressure prior construction: For intersection i and candidate green phase a, the raw pressure score is the net halting-queue pressure: the sum, over the phase's incoming–outgoing lane pairs, of (halting count on the incoming lane) − (halting count on the outgoing lane), taken from the last simulation step. Raw scores are normalized within each intersection:

~{p}_ia = clip((p_ia − mean_a p_ia) / max(SD_a(p), 1.0), −3, 3),

where the mean and standard deviation are taken over the phases of the same intersection and the standard deviation has a floor of 1.0 (pressures are vehicle counts); the standard deviation is the population SD over the intersection's phases (ddof = 0). Raw halting-queue imbalance is used rather than the capacity-aware refinement of PRCOL [10] because the prior's role is a relative preference among phases at decision time, not a throughput estimate. The IPPO-P policy logits are the baseline actor logits plus 2.25 × ~{p}_ia, with policy temperature 1.0 and no hold bonus for the current phase (the implemented hold-prior machinery carries a zero bonus in the frozen configuration). Three properties of this construction matter (Figure 1). Centering within an intersection removes the overall congestion level, so the prior encodes the relative pressure preference among phases rather than how busy the intersection is. Scaling by the within-intersection standard deviation (floor 1, because raw pressures are small-integer counts) keeps the logit perturbation commensurate across intersections and demand regimes without hand-tuned per-scenario constants. Clipping to [−3, 3] is an implementation safeguard that never activates in the configurations studied here: the standardization is taken over the n = 4 green phases of one intersection under the population-SD convention, so the maximum attainable standardized magnitude is $\sqrt{(n-1)} \approx 1.73$, and the effective worst-case logit shift is $\pm 2.25 \times \sqrt{3} \approx \pm 3.90$. We therefore do not attribute any binding role to the clip; a sensitivity protocol over the clip range and the scale remains future work (Section 6.7). One sensing asymmetry should be recorded: the prior consumes per-lane halting counts, a channel that is not part of the observation vector (which contains lane densities and the phase one-hot). Both actors receive the identical observation, but IPPO-P's action distribution depends on strictly more state information than IPPO's; the comparison evaluates the prior integration as a whole, this channel included.

4) Training-consistent PPO likelihood: The likelihood specification is given at tuple level (Figure 1). The on-policy buffer is appended at every environment step (one tuple per agent per 10 s step) and consumed by a PPO update after each episode; no decision-slot filtering is applied. Each tuple records the observation, the sampled action, the rollout log-probability under the prior-augmented distribution, the critic value, the step reward, the done flag, and the prior vector. Steps on which the signal is inside its minimum-green dwell or its yellow transition therefore also enter the buffer; on such steps the environment overrides the requested action (the current phase is kept or the yellow transition completes), while the tuple retains the agent's sampled action and its rollout log-probability; hence executed-action statistics differ from sampled-action statistics by construction. During the PPO update, the new log-probability of the stored action is recomputed from the stored observation, and the stored prior vector is reused verbatim; there is no recomputation of the prior from environment state at update time. The update therefore evaluates exactly the same

prior-augmented distribution family that generated the data, and the importance ratio exp(log π_new − log π_old) compares two members of that family. Scale and temperature are applied in the same order in both places (logits + $2.25 \cdot \sim\{\mathrm{p}\}$, then division by the temperature, 1.0 throughout), and the clip was applied once at prior construction, so rollout and update share the same clipped prior. Advantages are computed per episode with $\gamma = 0.99$ and $\lambda = 0.95$, bootstrapping with a zero value at the truncated final step and standardized per agent within the update. Because the stored prior is a deterministic function of the halting state at collection time and is itself stored, the modified policy remains a valid conditional distribution over phases at every stored tuple, and PPO's entropy bonus and clipped-ratio objective operate on it unchanged. The actor and critic are unchanged; the prior is an additive, input-dependent shift of the logits.

The consistency requirement defines the treatment. Biasing actions only at evaluation time would make the behavior policy differ from the policy that generated the stored old log-probabilities, invalidating the clipped importance ratio and changing what PPO estimates. Moreover, the prior also shapes which states the fleet visits during the 50 training episodes, so its influence extends beyond logit arithmetic at evaluation time. The comparison to IPPO therefore evaluates the complete training-consistent prior integration, not an isolated inference-time bias.

5) Training and evaluation protocol: Each task trains one method on one scenario with one training seed for 50 episodes; a single checkpoint is written after the final episode (final actor and critic weights plus configuration, with optimizer state kept only in the per-episode resume files), so evaluation uses the end-of-training policy with no best-episode selection. Evaluation reloads the checkpoint in fresh environments with learning disabled and deterministic action selection (the argmax of the same modified logits for IPPO-P, and of the baseline logits for IPPO), so the deployed control law is the trained prior-modified policy rather than a stochastic sampler. Stochastic-sampling evaluation was not assessed; the deterministic choice is symmetric across the two methods. Five evaluation episodes run under fixed evaluation seeds 1042–1046. These seeds enter the evaluation only through reseeding of the Python/NumPy/PyTorch generators and through the simulator's own seed (vehicle insertion and internal simulation randomness); with actions taken by argmax, an evaluation episode is a deterministic function of its seed and the checkpoint, and the evaluation seeds do not influence action sampling. Scenario bindings are given in Table 1; evaluation settings are identical across all tasks except for the scenario binding and the seed. The primary confirmatory population is ten new independent training seeds 47–56; pilot training seeds 42–46 are excluded from primary inference (Section 4.1).

6) Outcome measures: The primary outcome is demand service time U as defined in Section 3.1. Secondary service metrics logged over the same episodes are: mean travel time and delay of completed vehicles; lane-aggregated queue length (mean halting count × 7.5 m over unique controlled lanes), waiting time (mean accumulated waiting over unique vehicles on those lanes), and speed (vehicle-weighted mean over the lane union). These secondary metrics are computed over the same evaluation episodes as U, which makes them correlated with the primary outcome rather than independent replications of it. Only U and the operational audits were pre-specified for reporting (Section 4.5); the secondary service metrics were logged but excluded from the pre-registered analyses and are not reported or interpreted in this paper; no descriptive or inferential claim, and no scenario-specific test, is based on them. Completion bookkeeping retains two distinct denominators: planned_completion_rate = normal compliant arrivals / planned demand, and completion_rate = completed vehicles / departed vehicles. In the runs reported here the two coincide numerically, but they answer different questions (demand fulfillment versus fleet-level throughput) and would diverge whenever insertions fail or abnormal arrivals occur. The integrity audits have asymmetric epistemic status and are reported as such: the demand-conservation check is a bookkeeping identity (each planned vehicle is assigned exactly one demand state) satisfied by construction of the accounting routine, and it is not an independent physical-flow assertion (arrived + in-network = departed is not separately tested); the substantive hard gate is the zero-teleport requirement, which is enforced at aggregation and would invalidate the metric interpretation if violated. Completion-rate statistics provide the operational evidence that complements these checks.

D. Experimental Design and Statistical Analysis

1) Pre-registration and separation of pilot from confirmation: The confirmatory protocol (`ippo_prior_confirmation_v1`) was frozen on 2026-10-24, 22:46 (UTC+8), before any confirmatory data were generated. The registration is internal rather than registry-based: the frozen protocol file and its SHA-256 hash, together with the 2026-10-24 freeze timestamp, are author-controlled artifacts, and no external registry entry (e.g., OSF or AsPredicted) exists; the term "pre-registered" is used in this internal, hash-locked sense throughout the paper. The frozen file enumerates the training seeds, evaluation seeds, scenario bindings, hypotheses, the primary metric with its censoring rule, the primary contrast and test, and the method lock (prior scale 2.25, temperature 1.0, configuration `pressure_zscore_hold`); the remaining lock-referenced constants (the zero hold bonus, the standard-deviation convention and floor, the clip range, the signal phase structure, and the deterministic evaluation mode) are recoverable from the per-task manifests and the archived source files identified there, rather than restated in the frozen file itself. A preceding pilot four-grid study (Section 5.6) used training seeds 42–46; these seeds informed method selection and are excluded from

all primary inference. In particular, the method-lock constants (the pressure-prior scale 2.25, the within-intersection normalization with its standard-deviation floor, the clip range, and the zero hold bonus) were selected during the pilot phase and carried unchanged into the frozen confirmatory protocol. This is method selection informed by pilot data; it is compatible with confirmatory hygiene because the confirmatory data did not tune these constants and the confirmatory population consists of ten new independent training seeds 47–56. The frozen exclusion rule permits dropping a seed only for protocol-invalid or corrupted tasks, with the reason reported and the same seed rerun before aggregate results are unblinded; no such exclusion occurred. The stopping rule required all 60 planned tasks (3 scenarios × 10 seeds × 2 methods) to run, with no interim efficacy stopping.

2) Paired design and estimand: The training seed is the unit of analysis. Within each training seed and scenario, U is averaged over the five fixed evaluation seeds; the five evaluation episodes are nested repeated evaluations of the same trained policy, not independent training replicates. The per-seed effect is the equal-weight mean of log(U_IPPO / U_IPPO-P) over H1, H2, and H3. The aggregate effect is the mean of these ten seed-level log ratios, transformed to a percent improvement as $100 \times (1 - \exp(-\text{mean log ratio}))$. The log-ratio form is symmetric in the two methods and the percent transform reads directly as a relative reduction in U; equal scenario weights were fixed before data collection. Pairing by training seed is the design choice that matters most: run-to-run variance is a dominant uncertainty source in deep RL [16], [18], and seed-level pairing removes it from the method comparison rather than averaging over it. Reporting seed-level effects also keeps the sample size honest (ten seeds, not sixty tasks or three hundred episodes [17]).

3) Primary hypothesis test: The pre-registered primary contrast is IPPO-P versus IPPO on U under the estimand of Section 4.2. The test is an exact two-sided paired sign-flip test over the ten seed-level log ratios at $\alpha = 0.05$, with the decision rule "positive aggregate effect and exact two-sided $p < 0.05$". The null hypothesis is symmetry of the seed-level log-ratio distribution about zero. The statistic counts positive seed effects; under this null each seed's effect is equally likely to be positive or negative, so the null distribution is binomial over ten signs and the all-one-direction outcome yields $2 \times 2^{-10} = 0.001953125$ (reported as 0.001953 throughout the remainder of the paper). In statistic form this is the exact paired sign test; the frozen protocol names it a sign-flip test, and under the all-one-direction outcome observed here the sign test and the randomization over the 2^{10} sign assignments of the magnitudes yield the identical p-value, so the label and the reported value agree. The test involves no normal approximation and no variance estimate, which matters at this sample size. One primary contrast is pre-registered; no multiplicity correction applies to it. Exact permutation p-values are reported without zero-floor artifacts [20].

4) Uncertainty quantification: A paired percentile bootstrap over training seeds (20,000 resamples, RNG seed 20261024 [21]) provides 95% confidence intervals. Resampling over training seeds propagates the dominant variance source into the interval and matches the unit-of-analysis choice of Section 4.2. These intervals are descriptive; they do not replace the exact primary test and are not used for decisions. With ten seeds, percentile coverage is imperfect, which is one reason these intervals are labeled descriptive rather than inferential.

5) Secondary analyses: Secondary analyses are: (i) scenario-specific effects on the confirmation seeds, reported descriptively with bootstrap intervals but no per-scenario significance claims; (ii) a pooled 15-seed summary combining pilot and confirmation seeds, labeled descriptive only, since pilot seeds preceded and informed the frozen protocol and do not constitute independent replication of it; (iii) operational audits: completion rates under both denominators, teleport counts, and demand conservation. The secondary service metrics logged under Section 3.6 are not part of these analyses. No secondary outcome was designated as pivotal, no scenario-specific test is treated as confirmatory, and none of the secondary analyses can promote the result beyond the primary contrast's confirmatory scope.

6) Exploratory extensions: Three scale-up settings on the 3×3 grid were run after the confirmatory result was known: H1HIGH (18 route flows × 150 veh/h = 2700 veh/h total; n = 5 training seeds 62–66), H1SAT (a repository label carried for continuity, not a saturation finding; 18 × 300 = 5400 veh/h; n = 5, seeds 67–71), and H1PROBE700 (18 × 350 = 6300 veh/h; n = 2, seeds 72–73). All three are post-confirmation exploratory extensions: the frozen confirmation protocol was copied, not re-registered; none is merged into the primary test; none carries confirmatory status. For the probe, branch decision rules were locked during the runs, before any evaluation output was available, with n = 2 explicitly incapable of confirming any direction. Flow totals are counted from the 18 route-flow entries; nominal per-intersection demands of 300, 600, and 700 veh/h assume the nine signalized intersections as the denominator. Route-file comments contain inconsistent per-intersection figures, so labels such as "saturated" are not treated as evidence of network saturation. The 18 flows comprise 12 through routes and 6 turning routes, not an allocation balanced across movement classes. The probe additionally carried a pre-locked secondary guard: evidence of completion collapse or teleporting in either method would be treated as a protocol anomaly to be resolved before any mechanism interpretation. No guard condition was triggered.

7) Reproducibility: Task manifests (each task's `protocol_state.json` record) retain SHA-256 hashes of all source files, route-file bindings, variant configurations, checkpoint paths, and the deterministic fresh-environment evaluation mode; source-hash sets are consistent within each scenario across the 60 tasks. The manifest hashes cover the code, configuration, network, and route files that define each task, so a task can be re-derived from the manifest plus the

hashed sources. After the frozen runs, the source file `paper_protocol.py` was edited (the project is not under version control), so its current hash no longer matches the hashes recorded in the frozen manifests, while the other hashed components (environment, agents, scenario files) are unchanged. Bit-exact re-evaluations from stored checkpoints under the current code reproduce the stored per-episode metrics for each re-run task (three tasks across two audits: the two H2 pilot re-checks of Section 5.6 and the censoring-decomposition re-run of Section 6.6; deposit artifacts in the Data Availability statement), supporting functional equivalence of the evaluation path; the drift itself is disclosed rather than resolved. Training writes a checkpoint at episode completion, and evaluation reloads the checkpoint recorded in the manifest. Two classes of provenance were initially not captured at freeze time. Runtime and library versions and hardware descriptions were subsequently recovered from the training machine and are deposited with the archival package's environment snapshot: SUMO binary v1_25_0+0882 with TraCI/sumolib Python client packages 1.27.1, Python 3.9.25, PyTorch 2.8.0+cu126, dual Intel Xeon Gold 6230R with one NVIDIA RTX A6000. SHA-256 hashes of the checkpoint binaries were computed post hoc from the retained files and are deposited alongside. Git commit identifiers remain unrecoverable (the project is not under version control), and this is disclosed as a known gap (Section 6.6); the omission affects provenance bookkeeping only, not the logical protocol, which is fully specified by the frozen file and manifests.

E. Results

1) Protocol completion and integrity checks: All 60 planned tasks completed. Across the 300 evaluation episodes (60 tasks × 5 evaluation seeds), the demand-state bookkeeping check passed in 300/300 episodes (an identity satisfied by construction; Section 3.6) and no teleport events occurred in any episode, which is the substantive integrity gate. Planned completion ranged from 98.681% to 99.056%. In these runs, completion_rate and planned_completion_rate coincide numerically, but they retain distinct denominators (completed vehicles over departed vehicles versus normal compliant arrivals over planned demand) and are reported separately throughout. These 300 episodes are repeated evaluations nested within 60 training tasks (10 training seeds × 3 scenarios × 2 methods); they are not independent training replicates, and all inferential statements below treat the training seed as the unit.

Table 1. Scenario specification. Episodes: 50 training episodes per task; 5 deterministic evaluation episodes per task in fresh environments (evaluation seeds 1042–1046).

Scenario	Network	Signalized intersections	Route flows	Total demand (veh/h)	Planned vehicles per evaluation episode	Nominal per-intersection demand (veh/h)	Training seeds (primary)
H1	3×3 grid (`3x3.net.xml`)	9	18 (12 straight, 6 turning)	1800	5004	200	47–56
H2	2×2 grid (`2x2_oversaturated.net.xml`)	4	12 (8 straight, 4 turning)	2400	6672	600	47–56
H3	2×2 grid (`2x2_oversaturated.net.xml`)	4	12 (8 straight, 4 turning)	3000	8340	750	47–56

Notes: Totals are sums over route-flow entries of the route files; the nominal per-intersection demand divides the total by the number of signalized intersections and is an accounting quantity, not a measured saturation level. Network file names, including `2x2_oversaturated.net.xml`, are repository labels and are not treated as evidence of network saturation. Each signalized intersection runs an 8-phase signal program (4 green phases of 33 s plus 4 yellow transitions of 2 s); the controllable action space is the 4 green phases, the yellow interval being an automatic transition. Pilot training seeds 42–46 used the same three scenarios (Section 5.6). Planned-vehicle counts are observed per-episode values from the evaluation runs.

2) Primary confirmatory result: **Table 2. Confirmatory protocol and primary result.**

Item	Value
Estimand	IPPO-P vs IPPO on demand service time U (lower is better); per training seed: mean U over 5 evaluation seeds per scenario, then equal-weight mean log ratio over H1/H2/H3
Population	Ten new independent training seeds 47–56
Effect (percent transform)	**+1.712%**
Bootstrap 95% CI (descriptive)	[1.252%, 2.202%] (20,000 paired resamples, seed 20261024)
Seed direction	10/10 training-seed effects positive
Primary test	Exact two-sided paired sign-flip test, $\alpha = 0.05$, one internally pre-registered contrast, no multiplicity correction
p-value	**p = 0.001953**

Item	Value
Decision	Pre-registered hypothesis confirmed within the frozen scope

The ten seed-level log ratios range from 0.0050 to 0.0323 and are all positive (Figure 2). The sign-flip test's p-value is the smallest attainable with ten seeds, and it is reached because the direction is unanimous, not because any single seed shows a large margin; the smallest seed effect (0.0050, seed 52) would on its own be indistinguishable from noise. This is the intended reading of a sign test over paired seeds: the evidence is the consistency of direction across independent training runs, with the bootstrap interval describing magnitude. Descriptive scenario means of U over the confirmation seeds (n = 10 per scenario) are: H1, 135.15 s (IPPO) vs 134.60 s (IPPO-P); H2, 116.52 s vs 115.33 s; H3, 123.27 s vs 118.73 s. The effect is modest in magnitude and uniform in direction.

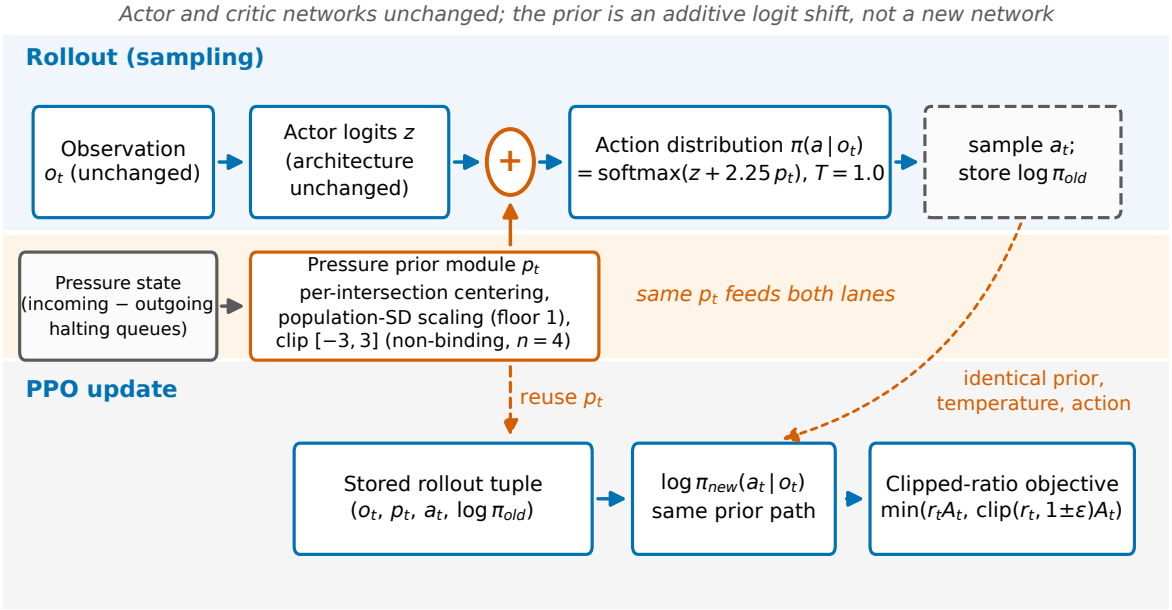


Fig. 1. **Figure 1. Method schematic.** IPPO-P computational and training-consistency diagram: observation → baseline actor logits; pressure state → per-intersection normalized and clipped prior (centered, SD-scaled with population SD and floor 1, clipped to [−3, 3], a bound that never binds at the 4-phase intersections); scaled addition (× 2.25) before the categorical distribution at temperature 1.0; the same stored prior and distribution used in rollout and in the PPO update for the stored old log-probability. The figure makes explicit that the actor/critic architecture is unchanged: the prior is an additive logit shift, not a new network. Source figure: `paper/figures/figure1_method_schematic`. *Data source:* Section 3.3–3.4 and the frozen method lock in `protocol_frozen.json`.

3) Evaluation-realization sensitivity (post hoc reanalysis): All ten training seeds are evaluated on the same five evaluation seeds (1042–1046), so the five episodes are shared realizations across seeds. The primary pairing and test are unaffected (the unit of analysis is the training seed), but the unanimity of the primary result could in principle depend on averaging over these shared realizations. A post hoc reanalysis therefore re-examines the direction at the level of single evaluation episodes, using the retained per-episode metrics (deposit: `results/eval_realization_sensitivity/`).

The episode-to-seed mapping is established from run artifacts rather than from the current source tree, whose post-freeze drift is disclosed in Section 4.7: for each of the 60 tasks, the per-episode rewards recorded in the run logs match the stored per-episode metric arrays in order (60/60 tasks). As a recomputation check, the five-episode means reproduce the stored per-task U exactly (maximum absolute difference 0), and the per-scenario and aggregate log ratios reproduce the stored values exactly.

For each evaluation episode e and training seed s, define the single-episode log ratio r(s, e) as the equal-weight mean over H1–H3 of log(U_IPPO(s, e) / U_IPPO-P(s, e)). An exact two-sided sign test over the ten training seeds within each episode gives:

Evaluation episode (seed)	Positive seeds	Exact two-sided p	Mean r(s, e)
e0 (1042)	10/10	0.001953	+0.016638
e1 (1043)	10/10	0.001953	+0.019539
e2 (1044)	10/10	0.001953	+0.016289
e3 (1045)	10/10	0.001953	+0.014825

Evaluation episode (seed)	Positive seeds	Exact two-sided p	Mean r(s, e)
e4 (1046)	9/10	0.021484	+0.018879

The direction is unanimous within four of the five evaluation realizations and 9/10 in the fifth; the single negative seed-episode is training seed 52 at evaluation seed 1046. Realization-level weakness exists at the scenario level (in H1 at evaluation seed 1042 only 3/10 training seeds are positive, p = 0.344, and H2 at 1045 splits 5/10), while H3 is positive in 9–10/10 seeds in every realization. The mean of the single-episode statistics (0.017234) lies 3.4×10^{-5} below the frozen primary statistic (0.017267), the Jensen gap between mean-of-logs and log-of-mean, so this reanalysis quantifies the sensitivity of the reported unanimity to realization sharing without revising the primary estimand. The five within-episode tests share training runs and scenario draws; they are descriptive decompositions, not independent replications or five confirmations.

4) Scenario-specific secondary results: **Table 3. Scenario-specific secondary effects on confirmation seeds (n = 10 per scenario; descriptive, not three independent confirmatory tests).**

Scenario	Effect (percent)	Mean log ratio	Bootstrap 95% CI (descriptive)	Positive seeds	U: IPPO (s)	U: IPPO-P (s)
H1	+0.407%	0.00408	[−0.048, 0.784]	8/10	135.1523	134.6031
H2	+1.023%	0.01028	[0.581, 1.560]	10/10	116.5230	115.3285
H3	+3.675%	0.03744	[2.845, 4.744]	10/10	123.2698	118.7265

These effects are secondary descriptions of the same confirmatory data, not separate hypothesis tests; no per-scenario significance claim is made, and the H1 interval crossing zero is reported as observed. The spread across scenarios (nearly flat in H1, largest in H3) motivates the boundary analysis of Section 5.5 rather than any load-response interpretation.

5) 5.5 Exploratory 3×3 scale-up and boundary results: **Table 4. Exploratory 3×3 results (all post-confirmation, exploratory; never merged with the primary test).**

Setting	Total demand (18 flows)	Nominal per intersection	n (training seeds)	Effect (percent)	Positive seeds	Exact sign-flip p (not interpreted)	Mean U: IPPO (s)	Mean U: IPPO-P (s)
H1HIGH	18 × 150 = 2700 veh/h	300 veh/h	5 (62–66)	+0.458%	5/5	0.0625	136.45	135.82
H1SAT	18 × 300 = 5400 veh/h	600 veh/h	5 (67–71)	−0.997%	0/5	0.0625	140.82	142.23
H1PROBE700	18 × 350 = 6300 veh/h	700 veh/h	2 (72–73)	+0.532% / +0.027% (per seed)	2/2	not applicable (descriptive probe)	144.72 / 143.08	143.95 / 143.04

Notes: All rows are post-confirmation exploratory extensions, as stated in the caption. Per-seed values for H1PROBE700 are listed in seed order (72, 73), separated by slashes. The exact p-values of 0.0625 are the minimum attainable with n = 5 and are shown for completeness only; as exploratory quantities they are not interpreted as significance evidence.

Operational checks: in H1HIGH and H1SAT all evaluation episodes passed demand conservation with zero teleports, and the minimum completion rate over episodes was ≥ 0.9864 (H1HIGH) and ≥ 0.9848 (H1SAT) for both methods. In H1PROBE700, conservation passed, teleports were zero, and minimum completion rates were ≥ 0.9848 for both methods.

At 2700 veh/h the direction matches the confirmation result (+0.458%, 5/5 positive, exact p = 0.0625, the minimum attainable with n = 5; descriptive only). At 5400 veh/h the effect reverses: −0.997% with 0/5 seeds positive. This is a relative difference under globally degraded conditions, not a method-specific operational failure: mean U rises by roughly 6–8 s relative to the 1800 veh/h confirmation scenario H1 for both methods, while completion remains high and teleport-free. The reversal therefore reflects a change in the relative value of the prior under load, not a breakdown of either controller. The 6300 veh/h probe shows paired improvement in both seeds, but one effect is only +0.027% (a near-tie), and n = 2 cannot confirm any direction.

No smooth or monotonic pressure-response curve is inferred from these points: topology, route composition, load, and sample size differ across settings, the flow totals are verified from the route entries (18 × 150/300/350 veh/h), the nominal per-intersection figures assume nine signalized intersections, and route-file labels are not treated as evidence of saturation. The substantive summary is a negative and a near-null point alongside the confirmatory positives: the confirmatory gain does not generalize unconditionally across topology-load settings.

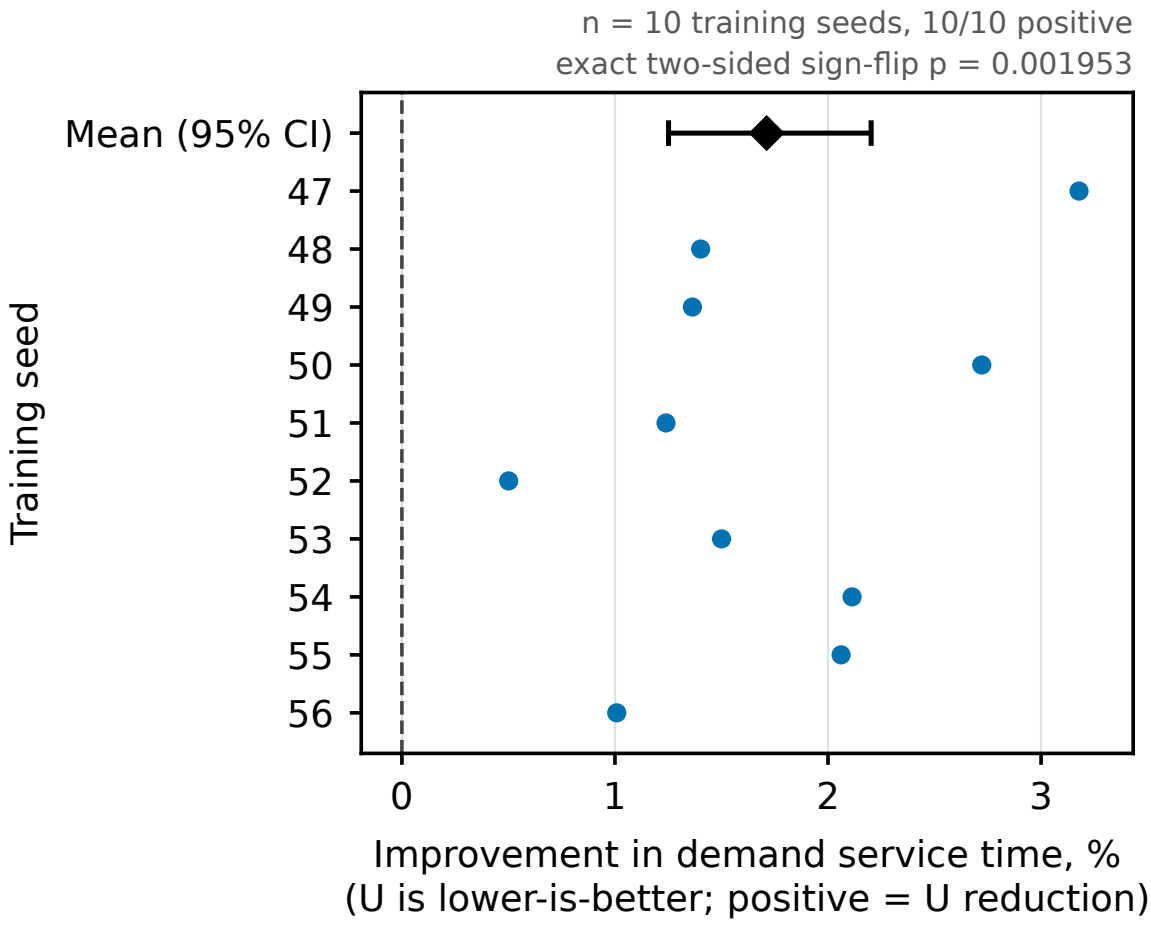


Fig. 2. **Figure 2. Seed-level effects.** Paired log-ratio effects of IPPO-P vs IPPO per training seed (47–56), zero reference line, aggregate estimate with descriptive bootstrap interval; n = 10 training seeds; exact two-sided p = 0.001953. The plotted quantities are raw log ratios; the percent transform of a log ratio r is 100 × (1 − exp(−r)), under which the aggregate point corresponds to +1.712%. Source figure: `paper/figures/figure2_seed_effects`. *Data source:* `seed_log_ratios` in `confirmation_summary.json` (0.0323, 0.0141, 0.0137, 0.0276, 0.0125, 0.0050, 0.0151, 0.0214, 0.0208, 0.0101 for seeds 47–56).

6) Pilot four-grid results (exploratory architecture context): The pilot study (training seeds 42–46, same H1/H2/H3 scenarios) preceded the frozen protocol and informed method selection. Its "four-grid" design is the 2 × 2 of architecture (independent PPO vs. a graph-communication variant, `GC`) × prior (absent vs. present); the two GC cells are identical to each other in every parameter except the prior scale (0.0 vs. 2.25), and the GC architecture as a whole changes multiple components simultaneously relative to IPPO (graph-residual independent actor, pressure observation, an action-value head, and soft-Q execution). The five pilot contrasts were: (i) the prior within IPPO (IPPO-P vs IPPO), +2.219% (5/5 positive, exact p = 0.0625); (ii) the prior within the GC architecture (GC-P vs GC-noP), +0.272% (3/5 positive, p = 0.5625); (iii) GC-noP vs IPPO (architecture effect without prior), +1.754% (5/5, p = 0.0625); (iv) GC-P vs IPPO (architecture effect with prior), +2.021% (5/5, p = 0.0625); and (v) the architecture-by-prior interaction, −1.991% (0/5 positive). After Holm correction across the five contrasts, none was significant (minimum Holm-adjusted p = 0.3125). Contrast (i), obtained before the confirmatory protocol was frozen, pointed in the same direction as the later confirmatory contrast and motivated the confirmation. Because the GC variants alter several components at once, no claim, positive or negative, is made about the independent value of graph communication. Pilot data never enter confirmatory inference; the pooled 15-seed effect of +1.881% (15/15 positive) is reported as descriptive only. A post hoc forensic recheck of the one anomaly flagged by a consistency audit (bit-identical five-episode evaluation vectors for the H2 pilot seeds 45 and 46) confirmed two independently trained checkpoints (distinct weights and distinct training curves) whose deterministic policies coincide on the shared evaluation trajectories; the pilot values stand as computed (deposit: `results/audit_recheck_v1/`).

7) External descriptive anchors (post hoc, not part of the confirmatory protocol): After the confirmatory analysis was complete, two non-learning anchors were run under the same evaluation setup. Fixed-Time replays the signal programs

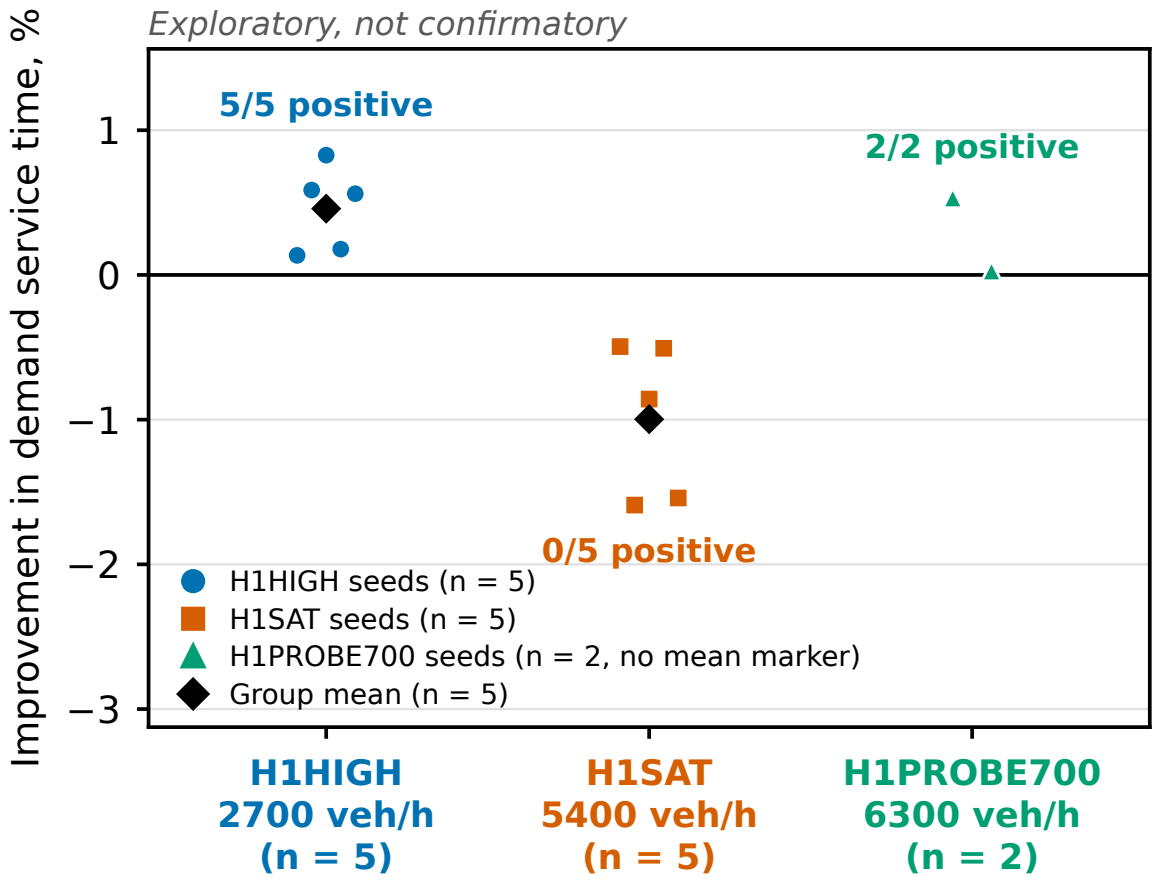


Fig. 3. **Figure 3. Exploratory boundary results.** Discrete effect points for H1HIGH (n = 5), H1SAT (n = 5), and H1PROBE700 (per-seed points, n = 2), with sample-size labels; the legend identifies the probe as per-seed points shown without a mean marker; scenario and demand annotated per point; no smoothed connecting line and no pooled inferential test. Source figure: `paper/figures/figure3_exploratory_boundary`. *Data source:* `scaleup_summary.json` (H1HIGH), `scaleup_v2_summary.json` (H1SAT), `decision_tree_locked.json` and per-task evaluation metrics (H1PROBE700).

embedded in the network files with no adaptive control (33 s green per phase plus 2 s yellow; observed cycles ≈ 156–172 s depending on the net). Max-Pressure-Descriptive is a Varaiya-style rule that, at each decision step, extends the phase with the largest total incoming-minus-outgoing queue pressure across controlled movements (minimum green 33 s; ties broken by program order). Both use the identical metrics path, demand window, and evaluation seeds 1042–1046, with no training. These anchors were constructed after the confirmatory analysis; they enter no statistical test, were not part of the pre-registered design, and do not alter any confirmatory claim.

Table 5. External descriptive anchors (n = 5 evaluation episodes per cell; post hoc, descriptive only). U SD is the sample standard deviation over the five evaluation episodes (ddof = 1); mean U for the learned arms is reported in Table 3.

Scenario	Controller	U (s, mean)	U (SD)	Planned completion (min)	Teleports	Conservation
H1	Fixed-Time	224.87	5.43	0.978	0	pass
H1	Max-Pressure	144.63	2.19	0.987	0	pass
H2	Fixed-Time	223.79	28.78	0.968	0	pass
H2	Max-Pressure	121.38	1.83	0.988	0	pass
H3	Fixed-Time	799.62	11.58	0.853	0	pass
H3	Max-Pressure	122.21	0.62	0.987	0	pass

Both learned policies lie well below the fixed-time anchor in every scenario. Against the max-pressure anchor, IPPO-P's descriptive mean U is lower in all three scenarios (134.60 vs 144.63; 115.33 vs 121.38; 118.73 vs 122.21; margins of 6.9%, 5.0%, and 2.9% of the anchor's U), while IPPO's is lower in H1 and H2 and higher in H3 (123.27 vs 122.21). With five evaluation episodes per cell and post hoc construction, these comparisons provide context on scale, not ranked

claims. The fixed-time breakdown under H3 (planned completion 85.3%, U ≈ 800 s) reflects an unadaptive program under the heaviest demand and is reported as anchor behavior, not as an algorithm comparison.

F. Discussion

1) Main finding: Within a protocol frozen before data collection, training-consistent integration of a normalized pressure prior into the PPO policy logits improved demand service time over an otherwise identical IPPO baseline: +1.712% with all ten training-seed effects positive and an exact two-sided p = 0.001953, the smallest p attainable with ten seeds. A post hoc decomposition by evaluation episode (Section 5.3) preserves the unanimous direction within four of the five shared evaluation realizations and 9/10 within the fifth, so the unanimity is not an artifact of averaging over the shared realizations. In absolute terms the descriptive scenario means imply per-vehicle savings of roughly 0.5 s in H1, 1.2 s in H2, and 4.5 s in H3, the latter across roughly 8,340 planned vehicles per evaluation episode. The remaining secondary service metrics (completed-vehicle travel time and delay, lane-aggregated queue, waiting, and speed) were logged but, as stated in Section 3.6, were excluded from the pre-registered secondary analyses; they are not reported here, their directions are not adjudicated, and they are not used as evidence of any kind, descriptive or confirmatory. The finding is confirmatory only within the frozen scenarios, metric, seeds, and analysis; it is not a claim about traffic signal control in general.

2) Scope of the evaluated treatment: The evaluated treatment is the complete training-consistent prior integration: the same modified distribution governs rollout sampling and the PPO likelihood evaluation through stored-prior reuse, preserving importance-ratio semantics (Section 3.4). This excludes two weaker interpretations. The result does not show that biasing a trained IPPO policy toward high-pressure phases at evaluation time would help; such a post-hoc intervention was not tested and would not be equivalent. Nor does it show that pressure belongs in the reward: the reward was identical across methods, so the effect isolates the prior's role in the action distribution. The scale (2.25) and normalization were selected during the pilot phase and fixed unchanged in the method lock (Section 4.1); no tuning over these values informs the confirmatory result.

3) Boundary evidence: The exploratory 3×3 results bound the claim. At nominal per-intersection demand of 600 veh/h, the effect is positive on the 2×2 grid (H2, +1.023%, confirmation seeds) and negative on the 3×3 grid (H1SAT, −0.997%, 0/5 seeds), while within the 3×3 family the effect moves from +0.41% at 200 veh/h (H1; confirmatory, n = 10) through +0.46% at 300 (H1HIGH; exploratory, n = 5), −1.00% at 600 (H1SAT; exploratory, n = 5), and a near-tie at 700 (H1PROBE700; exploratory probe, n = 2). These settings differ in topology, route composition, and load simultaneously, so the topology-load contrast is confounded and no factor attribution is attempted; the 12-through/6-turn composition of the 18 flows is held constant across the 3×3 settings, but that constancy does not make the load ladder balanced by movement class. What the data do establish is that the confirmatory gain is conditional: it can persist under increased load on one topology, reverse on another, and shrink to a near-tie at the highest tested load. The near-tie is easy to over-read: at 6300 veh/h the two seeds point the same way, but their margins (+0.532% and +0.027%) differ by an order of magnitude, so even the direction is not firmly established at this load. Universal-superiority claims are contradicted by the H1SAT result and are not made.

4) Mechanism: a hypothesis, not a finding: A variance-stabilization account (pressure biasing the sampling distribution toward queue-balancing phases, reducing destructive action variance during training) remains plausible but untested. The recorded data contain scenario-level aggregates, not intersection-level traces of pressure, chosen actions, or phase switches; with those traces absent, no causal mechanism is identified and none is claimed. The H1SAT degradation pattern (both methods worse in absolute U, IPPO-P relatively worse) and the near-tie at 6300 veh/h are consistent with competing bias-cost and stabilization-benefit terms whose balance shifts with operating conditions, but this reading is offered only as a hypothesis for future instrumented experiments. Two observable consequences would help discriminate it: seed-to-seed dispersion of U under the prior in regimes where its mean effect is positive, and intersection-level decision traces showing whether the prior disproportionately corrects queue-imbalanced decision slots. Such traces were not recorded in the runs reported here.

5) Relation to prior work: Pressure-informed RL controllers reported to date place pressure in the reward [9], [10] or the state [6], [11], largely with DQN-family value-based agents. The present work holds the reward and observations fixed and integrates the pressure signal additively into the logits of an on-policy PPO policy, applied consistently in rollout and update; in an on-policy method, whose sampling distribution is intrinsic to the learning rule, the policy logits are a natural integration point for such a prior. No head-to-head comparison against DQN-family pressure methods was run, so nothing is claimed about whether PPO-with-prior outperforms DQN-with-pressure-reward; the claims concern what the prior does to PPO under matched conditions. The contribution is consequently an evaluation of treatment consistency under an internally pre-registered protocol with seed-level reporting, rather than a new traffic-pressure principle or an architectural advance. Classical max-pressure control [8] and its queueing-theoretic origins [7] provide the conceptual source of the prior; no performance comparison against them is claimed here.

6) Limitations:

- One simulator family (SUMO) and synthetic grid configurations only; no real-world or field data.
- The external anchors (fixed-time program; descriptive max-pressure) were constructed after the confirmatory analysis and are reported as descriptive context only (Section 5.7); they follow documented fixed rules but were not part of the pre-registered design.
- Ten confirmatory training seeds; exploratory extensions use n = 5 and n = 2, the latter incapable of confirming any direction.
- Topology and demand are not factorially separated in the 3×3 extensions; the H2-versus-H1SAT comparison is confounded.
- The contribution of censored (incomplete) vehicles to the primary metric U is quantified by an archived decomposition on one representative confirmation task (H1 under IPPO, training seed 47; deposit `results/censoring_decomposition_v1/`): a bit-exact re-run of its five evaluation episodes, cross-checked against the simulator's per-vehicle tripinfo records, places the censored share of U mass at 0.60%–0.73% per episode (0.67% pooled over the five episodes), with censored vehicles 1.22% of planned demand (304 of 25,020), all of them still in the network at horizon end, and zero teleports. This is a single-task benchmark, not a per-scenario measurement; on it, normalizing over completed vehicles only (rather than censoring) would yield 136.69 s rather than 135.94 s. The censoring contribution is therefore small on the evidence available, but it is not measured per scenario.
- No intersection-level pressure, action, or phase-switch traces for the 5400/6300 veh/h runs, precluding mechanism identification.
- Pilot results informed method selection and remain separate from confirmatory inference; the pooled 15-seed summary is descriptive only. All hyperparameters, including the prior's constants, were carried over from the same-scenario pilot, so the confirmatory design removes confirmatory-data tuning but not the residual optimism that pilot-informed magnitudes carry (Sections 4.1, 6.2).
- The signal timing settings (a 2 s yellow interval, shorter than the 3–5 s typical of urban geometry, and a 10 s decision interval) are fixed and shared by both methods, but they were not varied, and the prior's interaction with different timing regimes is untested.
- Runtime/library versions, hardware descriptions, and checkpoint-file hashes were not captured in the frozen manifest; the first two have since been recovered from the training machine and deposited with the archival package's environment snapshot (SUMO v1_25_0+0882 binary, TraCI/sumolib client packages 1.27.1, Python 3.9.25, PyTorch 2.8.0+cu126, dual Xeon Gold 6230R + RTX A6000), and checkpoint-file SHA-256 hashes were computed post hoc from the retained binaries. Git commit identifiers remain unavailable (no version control) and are disclosed as such. In addition, the source file `paper_protocol.py` was edited after the frozen runs (the project is not under version control), so its current hash diverges from the manifest hashes; bit-exact checkpoint re-evaluations under the current code reproduce the stored metrics for each re-run task (Section 4.7), supporting functional equivalence of the evaluation path, but the drift itself is not reconstructible.
- No evidence of universal superiority, no real-world deployment, and no safety guarantees of any kind are implied.

7) Implications for future work: Two extensions follow directly, both as future pre-registered work rather than additional post hoc experiments here: factorial topology-by-load designs that separate the confounds in the 3×3 boundary results, and local trace logging (per-intersection pressure, sampled actions, and phase switches) sufficient to test the variance-stabilization hypothesis. A further contrast is a MAPPO arm carrying the same prior, which would test whether the effect survives a centralized critic [14]. A third extension is a sensitivity protocol for the prior's two free constants (the scale 2.25 and the clip range), evaluated as a pre-registered family of contrasts rather than tuned post hoc; the confirmatory result here covers only the frozen values. Scaling the seed count beyond ten would narrow the descriptive intervals and enable multiplicity-aware analysis of scenario-specific contrasts. For practitioners, the operational point is procedural: the prior's implemented cost is one addition per logit per decision step (an analytical characterization; no wall-clock or memory overhead was measured), changes nothing about the deployed architecture, and should be adopted on the basis of the same seed-paired, frozen-protocol evaluation used here rather than on the strength of any single-run pilot, including the pilots reported in this paper.

G. Conclusion

Under an internally pre-registered, hash-locked protocol frozen before data collection, this IPPO-internal controlled evaluation found that integrating a normalized, clipped pressure prior additively into the policy logits of independent PPO (identically during rollout and update) reduced demand service time relative to a matched IPPO baseline across three synthetic grid scenarios spanning two topologies: +1.712% (95% bootstrap CI [1.252%, 2.202%], descriptive), 10/10 training-seed effects positive, exact two-sided paired sign-flip p = 0.001953. The intervention is minimal: no change to reward, observations, architecture, or training budget; the prior is one addition to the logits, kept consistent between the behavior policy and the updated likelihood. This conclusion is confined to the frozen H1/H2/H3 scope and

to that specific treatment of training-consistent prior integration; it is a comparison of IPPO-P against IPPO alone and implies nothing about standing relative to external controllers or published benchmarks.

Post-confirmation exploratory evidence cautions against universal extrapolation. The same intervention showed a positive descriptive effect at 2700 veh/h on the 3×3 grid, a consistent negative effect at 5400 veh/h, and an inconclusive near-tie probe at 6300 veh/h; no pressure threshold is established, no variance-stabilization mechanism is proven, and no general superiority over existing traffic-control approaches is claimed. The confirmatory evidence covers a single point in the prior's design space (scale 2.25 at temperature 1.0); sensitivity to these constants is untested here. For scale context, both learned policies sit well below a fixed-time anchor in every scenario, and IPPO-P's descriptive mean U is 2.9%–6.9% below a max-pressure rule that in turn edges IPPO in the heaviest scenario; these figures give context on scale, not ranked claims. The intervention is reported exactly as evaluated (one frozen method, one metric, ten seeds, three scenarios), and its boundaries are reported with equal precision. For practitioners and researchers, the summary is narrow: training-consistent pressure priors are an architecturally minimal modification (one addition per logit at decision steps, with no change to the deployed network) that helped in every confirmatory seed tested here, and whose benefit is bounded by operating setting in ways that future factorial, instrumented studies should characterize.

H. Declarations

1) Data Availability Statement: All data supporting the reported results were generated by the frozen protocol `ippo_prior_confirmation_v1` and its documented exploratory extensions. The repository contains: the frozen protocol file (`results/ippo_prior_confirmation_v1/protocol_frozen.json`) and its SHA-256 hash file (`results/ippo_prior_confirmation_v1/protocol_frozen.sha256`); the confirmation summary with all seed-level effects (`confirmation_summary.json`); per-task manifests recording source-file SHA-256 hashes, route bindings, variant configurations, and checkpoint paths; per-task training logs, checkpoints, and per-evaluation-seed metrics; scenario network and route files (`nets/grid-3x3/`, `nets/2x2-grid/`); the exploratory outputs (`results/ippo_prior_scaleup_v1/`, `results/ippo_prior_scaleup_v2/`, `results/ippo_prior_probe_700/`, `results/formal_four_grid_v1/`); the post-hoc external anchors (`results/baselines_descriptive_v1/`, generated by `agents/baselines_descriptive.py` via `tools/run_baselines_descriptive_v1.sh`); and the three post-review audit/reanalysis deposits with their generating scripts: `results/audit_recheck_v1/` (pilot-duplicate forensic recheck and bit-exact re-evaluations), `results/eval_realization_sensitivity/` (per-evaluation-episode reanalysis of Section 5.3), and `results/censoring_decomposition_v1/` (tripinfo-verified censoring decomposition of Section 6.6). The frozen method lock records the pressure-prior construction under the configuration identifier `pressure_zscore_hold`. Runtime and library versions, hardware descriptions, and post-hoc SHA-256 hashes of the checkpoint binaries are deposited with the archival package (environment snapshot and `checkpoint_hashes.sha256`); git commit identifiers remain unavailable (the project is not under version control). [**FILL: public archival DOI/repository (e.g., Zenodo) — add upon deposit or acceptance.**]

2) Ethics Declaration: This study uses only synthetic microscopic traffic simulation. No human participants, human data, or animals were involved, and no ethics-committee approval is required.

3) Author Contributions (CRediT): **Rongtao Zhu:** Conceptualization, Methodology, Software, Investigation, Data Curation, Formal Analysis, Visualization, Writing – Original Draft. **Zhaoxia He:** Methodology, Validation, Formal Analysis, Writing – Review & Editing. **An Zhao:** Software, Investigation, Data Curation. **Shubin Wei:** Investigation, Visualization, Writing – Review & Editing. **Qiran Li:** Investigation, Data Curation, Writing – Review & Editing.

All authors read and approved the final manuscript and take full responsibility for its contents.

4) Conflict of Interest: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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6) Use of Artificial Intelligence: Artificial-intelligence tools assisted data aggregation, statistical checking, and manuscript drafting under direct human direction, following the frozen protocol and its pre-specified analyses. Every quantitative result in this paper is traceable to the frozen protocol outputs listed in the Data Availability Statement; no AI-generated numbers were introduced. The authors reviewed, verified, and revised all content and take full responsibility for the manuscript.

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